

Manan Mrig

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ML Engineer focused on building intelligent, scalable applications and the distributed systems that power them.

EDUCATION

University of Waterloo

Waterloo, ON

Master's of Data Science and Artificial Intelligence

Sep 2026 - Dec 2027

- Notable Coursework: Statistical Learning - Classification, Data Engineering, Machine Learning

University of Minnesota, Twin Cities

Minneapolis, MN

Bachelor's of Science, Computer Science

Sep 2020 - May 2024

- Notable Coursework: Machine Learning, Algorithms & Data Structures, Distributed Systems, Operating Systems

WORK EXPERIENCE

Software Engineer (Machine Learning Infrastructure)

June 2024 - July 2025

Target Corporation - Demand Forecasting Engine, Inventory Insights Acquisition

Minneapolis, MN

- Maintained and enhanced production GAMM-based forecasting pipelines for 100K+ SKUs using multi-year retail time-series data on Hadoop, supporting item- and chain-level inventory forecasting.
- Developed batch jobs and Kotlin Micronaut and Springboot microservices to enable human overrides of production forecasting models, giving planners greater control over inventory planning and physical inventory workflows.
- Owned production microservices supporting data-driven inventory interventions across 1,900+ stores, improving inventory reliability and system accuracy by 20% through ongoing enhancements and maintenance.

Teaching Assistant

January 2022 – May 2024

University of Minnesota, Twin Cities

Minneapolis, MN

- Led lab sections and office hours for Machine Architecture, Software Design for Data Scientists, Programming Concepts, and Discrete Mathematics, reinforcing concepts in operating systems, computer architecture, and systems programming.
- Designed, validated, published, and graded programming assignments in C, Python, Java, and Assembly while mentoring students through debugging and core computer science concepts.

Software Engineer Intern

June 2023 – August 2023

Target Corporation - Paystack

Minneapolis, MN

- Built Kotlin microservices for high-volume real-time payment processing, supporting low-latency analytics, monitoring, and reliable production payment workflows.
- Improved system performance by 40% through comprehensive unit testing and performance optimization, increasing throughput under production workloads.

Software Engineer Intern

May 2021 – July 2021

Centrl Inc.

Mountain View, CA

- Migrated backend components of a legacy PHP customer administration platform to Java and Spring Boot, reducing request processing time by 30%.
- Implemented OAuth2 authentication using Google's Authorization Framework in a Spring Boot application, streamlining secure access for 50+ internal users.

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, SQL, C++, JavaScript, TypeScript, C, Bash

Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Statistical Modeling, Matplotlib

Backend: Spring Boot, Micronaut, Flask, LangChain, Java RMI

Infrastructure: Docker, Kafka, PostgreSQL, Hadoop, Hive, AWS, GCP, Git, Vercel, Jira

PROJECTS

Personal AI Assistant Ecosystem | *SwiftUI, Kotlin, Python, RunPod, LangChain, & Postgres*

2026

- Developed a multimodal AI assistant using SwiftUI, Kotlin, Python, LangChain, and RunPod-hosted LLMs, implementing long-term vector memory and reducing inference latency by 40%.

Pro-Posterous: Human-AI Co-Creation Canvas | *Next.js, tldraw, LangChain, Diffusion Models*

2026

- Built an AI-powered poster design platform integrating diffusion models (FLUX, WAN, SDXL) with an interactive canvas for real-time iterative image editing.